



An optimal inflation rate for South Africa

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Abstract

The objective of this paper is to consider whether the inflation target of the South African Reserve Bank, which is currently between 3 and 6%, should be lowered to something closer to that of South Africa's trading partners, which is close to 3%. The paper reviews the theoretical literature and finds a strong contrast between much of the literature which argues for a zero inflation target and the practical implementation of inflation targeting around the world, where no country has adopted a zero target. Given this conflict it is clear that the theoretical literature is often too simplistic. The paper then considers the main arguments for a positive inflation target. The broad conclusion is that most of the arguments for a non-zero inflation target suggest a target in the range of 2–3%. However, any change in the target rate should be carried out slowly with full transparency and without loss of credibility.

Keywords Optimal inflation target · South Africa · Inflation

1 Introduction

This paper is part of the United Nations University World Institute for Development Economics Research (UNU-WIDER) research programme entitled 'Southern Africa—Towards Inclusive Economic Development'. The paper is part of the 'Macro-fiscal and analysis and modelling' workstream. The objective of the paper is to consider whether the inflation target of the South African Reserve Bank, which is currently between 3 and 6%, should be lowered to something closer to that of South Africa's trading partners.

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Since the seminal contributions of Kydland and Prescott (1977) and Barro and Gordon (1983), it has been understood that policy makers who are free to exercise full discretion in the setting of interest rates are unable to commit themselves to a low inflation regime. This is sometimes referred to as the problem of inflation bias, which stems from a lack of credibility on the part of policy makers as, in the face of a negative output shock, policy makers may induce inflation to offset the shock (Romer 1996). The result is the well-known time inconsistency problem and the loss of credibility to the policy maker. The solution to this problem is for governments to delegate monetary policy to an independent agency, normally the central bank, which has a clear and binding remit to set policy in such a way as to maintain a stable and low level of inflation. Under this regime, it is expected that the central bank will gain credibility and hence be able to directly affect inflation expectations and thereby control inflation at a lower cost than might otherwise be the case.

Over the last 30 years, the system of delegating monetary policy to the central bank, which has a clear legal remit to maintain inflation at a set target, has spread around the world and has become the likely dominant system of monetary governance. In 1990 New Zealand was the first country to adopt inflation targeting, followed by Canada in 1991 and the UK in 1992. Inflation targeting then spread rapidly. Appendix B lists 73 countries which currently use inflation targeting as their main approach for setting monetary policy and shows their chosen target level for inflation. These targets vary quite widely—most developed economies have chosen a target either at or close to 2% per annum. Many developing countries have chosen much higher target inflation rates—Liberia has a target of 8.5% with a band of 2% around this central rate, which would allow inflation of 10.5% before action is required. Inflation targeting is the longest-lasting policy system since World War II and, unlike other monetary systems, it has proved to be robust in the sense that, so far, no country has abandoned the approach. Walsh (2009) concluded his study by stating that inflation targeting is both feasible and sustainable.

King (1996, 1998) made the forceful point that the most important change in monetary policy was not inflation targeting itself but the realization that transparency and accountability are key to an effective monetary policy. He contrasted the monetary policy regime of the 1970 s and 80 s, which he described ‘as at best opaque’, with the later developments of a clear and transparent monetary regime. Given a clear and transparent framework, the details are far less important, ‘be it monetary targeting, the ERM [Exchange Rate Mechanism] or direct inflation targeting [they] have often coincided with sustained falls in inflation’ (King 1996).

It is surprising that, although a vast amount of literature has discussed the best type of regime for monetary policy and the consensus has clearly gone towards inflation targeting, there is very little academic consensus as to what that the target should actually be. Diercks (2019) surveyed almost all the academic studies which address this question. Of the 161 papers he listed, the recommended optimal target ranged between -7.6% and $+6\%$, with the vast majority of papers suggesting either a zero rate or a negative one. Similarly, there is little good analysis of how wide the bands should be or what time horizon should be set to meet the target after a shock has moved inflation away from its target. These are all key operational issues which any monetary authority faces.

This paper sets out to discuss these important practical questions and, in particular, to ask whether there is a case for lowering the current inflation target for South Africa, which is 3%–6%, to something closer to that set in the developed world.

The structure of the paper is as follows. Section 2 considers some key variables for South Africa, inflation, and the central bank interest rate and shows how the volatility of these variables has varied over time. Section 3 then considers the various arguments for how we might choose an actual inflation target. Here there is a sharp dichotomy between much of the literature and the practice of central banks. Much of the literature has concluded that a zero, or even a negative target, is optimal, while no central bank has so far adopted a zero inflation target. This section then asks how this sharp contrast has arisen and, in particular, discusses some recent papers which have given more support to the central bank's view. Section 4 then considers some operational issues in changing the inflation target, and Sect. 5 draws some general conclusions.

2 The data

A number of the arguments which are discussed in Sect. 3 regarding an optimal inflation target rest on measures of the volatility of inflation and interest rates. It is therefore useful to examine the data before discussing the theories so that we have a clear idea of the volatility of inflation and interest rates and how this has changed over time. These data and the analysis along with details of the data sources are presented in Appendix A. Figures 1 and 2 show the time series for annual consumer prices index (CPI) inflation and the central bank interest rate from 1959 month 1 to 2024 month 5, respectively. There are at least three quite distinct periods. From the beginning of the sample through to 1990, there was a fairly steady rise in inflation and interest rates, peaking at just over 20% a year for both. There was then a steady fall in inflation until the early 2000s when interest rates peaked again just before 2000. During 2005 inflation actually turned negative for a short period. This seems to have been caused by a period of industrial unrest with a number of strikes across the country, including a major strike in the gold mining sector which actually closed the mines for a period and caused considerable distortion to the measured data. There then seems to have been some instability around the time of the 2008 financial crisis and since then inflation has stabilized close to 5%, which is within the central bank target range, and interest rates have fluctuated but remained reasonably stable through to the end of the period.

For some of the models we discuss below, the natural real interest rate is a key variable. The natural real interest rate is defined as the real interest rate which neither increases nor decreases inflation, and it is normally assumed to be the real interest rate which emerges when the economy is in a steady state. The actual real interest rate along with a Hodrick–Prescott (HP) trend is shown in Fig. 3. This has clearly varied considerably over time but the average over the period after 2005 is 1.3%, which seems to be a reasonable result for the natural real interest rate over this

period, although the actual rate has varied quite considerably from a minimum of -3.8% to a maximum of 5.7% .

For completeness Table 1 gives the standard summary statistics based on the whole sample. This is not particularly informative as the basic properties of the series have clearly changed over time. The fact that the standard deviation of inflation is reported as 4.7 percentage points for the whole sample, which would mean we could easily expect inflation to vary by up to 9 percentage points (that is, roughly 2 standard deviations) from its target, is clearly not a relevant statistic for the period after about 2005. To define a more sensible set of measures of the likely variation in inflation and interest rates, we therefore perform two pieces of analysis for each series.

First, we build a standard autoregressive (AR) regression model to form a simple forecasting model where we start from an AR(12) model (as we have monthly data) and then nest down on the lag structure by eliminating insignificant lags. The absolute residuals from this model are then calculated as an instantaneous measure of the standard error of inflation at each point in time. Next, we estimate a standard generalized autoregressive conditional heteroskedasticity (GARCH(1,1)) model based on the same AR model and gather the conditional standard error as an alternative measure of the instantaneous volatility of each series. Tables 2 and 3 show the ordinary least squares (OLS)-based AR models for inflation and the interest rate, respectively, and Tables 4 and 5 show the corresponding GARCH(1,1) models. There are of course many possible models we could use here, for example which include such variables as house prices, unemployment, and interest rates. However, the objective here is not to find a perfect model but to calibrate the likely variation which might take place in inflation and interest rates. It is better to overestimate the size of this standard deviation by using a very simple model as this will give us a more conservative set of conclusions when we draw on the numbers below.

Finally, Figs. 6 and 7 show both the absolute errors and the GARCH(1,1) conditional standard errors first for inflation and then interest rates. This is an unusual and interesting comparison as both show a very similar pattern of volatility. The time variation in these measures is very obvious, but the key thing from the perspective of this paper is that since approximately 2008 the standard error in inflation has been very close to 0.5 percentage points and the standard error in interest rates has been even lower at around 0.25 percentage points except for a brief period in 2020, which was obviously a result of the COVID-19 pandemic. This will have important implications for the optimal inflation target when we come to consider the issue of the zero lower bound on interest rates, which is discussed below.

3 Arguments regarding the optimal inflation target

3.1 The Friedman rule

In theoretical models which include money and where the only nominal rigidity comes from the demand for money itself, the optimal monetary policy requires minimizing the opportunity cost of holding money by setting the nominal interest

rate to zero (Ireland 2003). The intuition behind this conclusion is that any positive rate of interest will induce individual economic agents to hold less money than would be optimal given the assumption that money can be created without any cost. The conclusion therefore is that the nominal interest rate should be zero and any inflation target should be consistent with this. This is known as the Friedman rule (Friedman 1969), and it has the implausible result that if the real interest rate were, say, 2% then the optimal rate of inflation would be -2% . This implies that the only sensible solution is for nominal interest rates, real interest rates, and inflation to be zero. There is, however, no particular reason why the natural real interest rate should be zero, as this would normally be assumed to come from the real economy as a whole and so it is usually assumed that the Friedman rule implies an optimal negative rate of inflation.

There is, in fact, considerable support for an inflation target of zero in the academic literature. Diercks (2017) surveyed all the literature from the mid-1990s to 2017 which derived an optimal inflation target. His analysis showed that the majority of studies produced either a zero or negative optimal inflation target, although the studies suggesting a positive target tended to be the more recent ones.

This is clearly at odds with the practice of all central banks which target inflation (see list of 73 countries which target inflation and their targets in Appendix B), as there is no central bank in the world which targets zero or negative inflation. Clearly, the simple theoretical models which underpin the Friedman rule are totally at odds with the expert practitioners who set the real-world targets. In this circumstance, we need to ask if the theoretical models, especially in the earlier parts of this literature, made a number of implausible assumptions, and the recent literature has examined these in some detail.

There are a number of important assumptions which, when relaxed, can give rise to a positive inflation target. These include: (1) the employment and output effects resulting from downward nominal price rigidities to an economy's ability to adjust relative prices; (2) the problem of the zero lower bound on interest rate policy; (3) the implications for servicing government debt and the problem of an incomplete tax system; and (4) varying relative productivity of individual firms over time. The following sections examine the arguments for an optimal positive rate of inflation and, where possible, discuss their relevance to the South African economy.

3.2 Benefits to growth of low inflation

There is a long literature on the benefits from low inflation (Barro 1996; Fischer 1993; Hess and Morris 1996) to gross domestic product (GDP) growth. Barro (1996), for example, suggests that a fall in inflation from 5% to zero might increase GDP growth by 0.1–0.5%. These estimates are largely derived from simple empirical analysis, but they are justified by appealing to the uncertainty which high and variable inflation brings. This is undoubtedly an important argument in general, and few would doubt that the degree of uncertainty induced by high and variable inflation would have seriously detrimental effects. However, this relationship is unlikely to be linear (as Barro 1996 assumes), as can easily be seen by considering the detrimental

effects of high and variable negative inflation. It is generally accepted that deflation reduces social welfare. If people have fixed nominal liabilities such as mortgages, then deflation will increase the real value of these liabilities and reallocate resources in a sub-optimal way. Barro's (1996) linear model would imply a continuing benefit as inflation becomes negative. Indeed, the data which Barro used have almost no observations with zero or even near zero inflation, and so it is likely that a reduction from say 2% to zero would not have the same effect as a reduction from 5 to 3% and could actually be much less beneficial. It is also almost certainly true that stable inflation at any level will have fewer detrimental effects on GDP than highly variable inflation.

3.3 Downward nominal rigidity

A key early paper in this area is the one by Akerlof et al. (1996), who developed a model with temporary downward nominal wage rigidities, monopolistic competition, and demand and supply shocks. This model generates a Phillips curve which is vertical at relatively high rates of inflation but where, at low rates, the curve becomes negatively sloped and can generate higher unemployment and hence lower growth. Their model was calibrated, and hence the results are not well founded in data, but it suggested a significant trade-off at very low rates of inflation, with zero inflation raising unemployment by around 2.5% and consequently leading to a significant loss in GDP. These findings were challenged by Howitt (1997) and Hogan (1997), who claimed the costs of firing and hiring workers as a reason why this may not happen, and by Bernanke and Parkinson (1991), who got a similar effect by appealing to labour hoarding. Woodford (2003) also argued for a zero inflation rate on the grounds that this would minimize distortions in relative prices.

The Akerlof et al. (1996) argument is sometimes referred to as greasing the wheels of the economy with a little inflation, thereby allowing relative prices and wages to adjust to a shock. The counter-argument is sometimes referred to as sand in the wheels due to the uncertainty caused by higher and variable inflation. The general consensus is that above around 5% inflation the sand effect dominates and below that the grease effect dominates (Card and Hyslop 1996; Groshen and Schweitzer 1997). But again, this relationship is not linear and downward nominal rigidities seem to be best coped with at an inflation rate of 2–3%.

The importance of downward rigidities was recently analysed in a much more sophisticated way by Adam et al. (2021), Adam et al. (2022), and Adam and Weber (2023), who based their analysis on some newly available data for the euro area on micro price data from the official CPIs for Germany, France, and Italy. These data are available through the Eurosystems Price-setting Microdata Analysis Network (PRISMA) and consists of over 80 million price observations over the period from 2010 to 2019. This allowed them to analyse trends in relative prices and to estimate the macroeconomic costs of sub-optimal inflation. Their broad findings were that the optimal inflation rate was between 1.1 and 2.1% for France, 1.2 and 2.0% for Germany, and 0.8 and 1.0% for Italy. The intuition behind this finding was that when a firm launches a new product, this product will normally be an innovation

and attract a relatively high price. Over time the product will become less desirable as new versions come to market, and its relative price should fall. A moderate rate of inflation allows this relative price adjustment more easily than actual price reductions.

An important part of the analyses by Adam and colleagues was that the optimal rate of inflation varies with different sectors of the economy. So, in the case of the food sector in France, optimal inflation was only 0.2%, as food prices were relatively flexible downwards, while non-energy industrial goods had an optimal inflation rate of 4.9% and services only 0.1%. Again, the intuition here is straightforward. In the food sector there is little need for relative price adjustments, in the industrial goods sector there is much more need, and in the service sector there is again little need and prices are less rigid. Similar results held for Germany and Italy. Optimal food inflation in Germany was -0.1% and was 0% in Italy. For the industrial sector it was 5.5 and 2.6% and for services it was -0.9 and -0.1% . These rates seemed to be quite stable over time, although their total sample was quite short. Their study is particularly important as it is the first to empirically estimate optimal inflation rates based on individual pricing decisions.

One important caveat here is that Adam and colleagues focused essentially on retail prices and did not allow for nominal wage rigidities, which many analysts would consider to be the most important nominal rigidity. Therefore, the very low inflation rates for the agricultural and service sectors may be unreasonably low.

Data such as this are not widely available for other countries and it is not clear how this would transfer to a country such as South Africa. The broad distribution of sectoral output for Germany is roughly 70% services, 29% industry, and 1% agriculture. As the equivalent for South Africa is 62% services, 24% industry, and 2.6% agriculture, this argument would suggest, if anything, a slightly lower target rate of inflation for South Africa as the industrial sector is slightly smaller but, again, this ignores nominal wage rigidities.

3.4 Government debt servicing and an incomplete tax system

Historically, inflation is probably the main way in which large government debt stocks have been corrected. After World War II the UK debt stock-to-GDP ratio was at a very high level of 250%, which was slowly eroded by inflation until it stabilized in the early 1980 s below 50%, largely due to the relatively high inflation of the 1970 s and 80 s. It is also true that a long-run change in the level of inflation from say 5% to 2% will imply a considerable transfer of wealth from the central government to bond holders, as they will be repaid more in real terms than they would have originally expected.

There are a number of ways in which an incomplete tax system can generate a positive optimal inflation rate. These include:

1. Untaxed profits due to decreasing returns to scale
2. Untaxed profits due to monopolistic competition in product markets
3. Untaxed income due to tax evasion.

All three forms of tax evasion can be at least partly compensated for by a general inflation tax. A positive rate of inflation means that the money supply and the government debt stock can both be expanded without increasing either the real stock of money or the debt-to-GDP ratio. These then allow government expenditure without increasing actual taxation.

Schmitt-Grohe and Uribe (2011) evaluated these three possibilities and concluded that a very large share (above 30%) of untaxed total income is required to justify an inflation rate of 2%. When we consider the range of targets in Appendix A, most developed countries are close to a 2% target, and we would not find it credible that tax evasion in these countries would be as high as 30%. Most developing countries have chosen a target well above 2%, and so this argument would require much higher rates of tax evasion.

A further distortion comes from a tax system which is not fully indexed, which is sometimes called fiscal drag. In most countries income tax allowance rates are set in nominal terms, as are various duties on such things as alcohol, fuel, and tobacco. A non-zero rate of inflation will require these to be regularly updated or clear distortions will occur, which can be to the benefit of the treasury in the case of income tax allowances and to its detriment in the case of duties.

Another possible omission is where a country's currency is widely used abroad. By having a positive inflation rate the country can effectively tax residents of other countries using its currency. This probably only applies to the USA, where the majority of US dollars are held abroad. However, it would have little relevance for South Africa and so we will not pursue this argument further.

It is of course hard to estimate the size of the informal economy and tax evasion, but it seems reasonable to assume that it is larger in South Africa than in most fully developed economies. So, these arguments could be seen as an argument for a somewhat higher target inflation rate in South Africa, and it may well be true that this is the reason why many developing economies have opted for a higher target, as shown in Appendix B.

It can also be argued that these are matters for the fiscal authority and that such decisions should be made at that level. An important factor here is that an inflation tax is generally a regressive tax (as the tax rises, the rich pay proportionately less and the poor proportionately more). The rich can protect much of their wealth and income by investing in property, businesses, and financial instruments such as index-linked bonds or shares, while the poor are unable to do this. This may be contrary to the wishes and objectives of the fiscal authority.

3.5 The zero lower bound

The use of interest rates as a stabilization tool becomes problematic when inflation rates are low or even negative and there is a need to stimulate the economy. Stimulating the economy would conventionally require a reduction in interest rates and, while we have seen some instances of negative interest rates since the 2008 financial crises, the degree to which interest rates can go negative is very limited. A higher target inflation rate will therefore protect central banks' ability to run

an effective stabilization policy. The relevance of this argument for maintaining a positive inflation target therefore rests on the volatility of inflation, interest rates, and GDP growth.

A seminal contribution in this area is the study by Andrade et al. (2019). An earlier, slightly less general study along similar lines was conducted by Coibion et al. (2012) and found broadly similar results, and so it will not be discussed further here. Andrade et al. (2019) start from a stylized fact that in advanced economies there has been a long-term decline in the natural real rate of interest (Holston et al. 2017; Laubach and Williams 2016), which may be due to a decline in the trend growth of productivity, demographic factors, or increased risk aversion in financial markets. The consequence of a lower real interest rate is that, given a set inflation rate, the nominal interest rate which is used for stabilization purposes will hit its zero lower bound more frequently. This could hamper counter-cyclical monetary policy and could potentially lead to more protracted recessions. This then argues for a higher target inflation rate and a smaller incidence of zero lower bound events happening but at the cost of the higher inflation rate. There is therefore a trade-off to be made between the cost of inflation and the cost of the zero lower bound events. This trade-off is also a function of the natural real rate of interest; a lower real rate of interest should lead to a higher optimal target rate of inflation. This of course requires a fairly complex model to evaluate this trade-off.

Andrade et al. (2019) investigated this issue by using an estimated new Keynesian dynamic stochastic general equilibrium (DSGE) model, which was estimated using US data. This framework included wage and price stickiness, so that low inflation is costly because of the difficulty in adjusting relative prices and real wages. It also featured a zero lower bound on the nominal interest rate. This model therefore contained the two main reasons (wage stickiness and the zero lower bound) for having a positive rate of inflation. The model was estimated using Bayesian techniques and was then simulated stochastically. Their analysis also allowed for parameter uncertainty, which makes a small difference to the optimal inflation target. The broad finding was that for the USA the optimal inflation target varies between 1.85 and 2.21% depending on the uncertainty of the parameters of the model. The probability of hitting the zero lower bound with this inflation target is around 6%. This is based on a real interest rate of around 2%. With a real interest rate of around 1%, the optimal inflation target would rise to around 3.5%. As the real interest rate rises, the optimal inflation rate falls and becomes effectively zero as the natural real interest rate rises above 5.5%

A related paper to the one by Andrade et al. (2019) is that by Blanco (2021), which varies from Andrade et al. (2019) in that the pricing system is much more detailed and involves a micro set of firms with a menu cost structure. Apart from this, the analysis follows a similar DSGE framework. The shocks are applied at the micro level to individual firms' prices and hence an overall misalignment of relative prices does not occur to the same extent. An important assumption is that, while prices are sticky, wages are fully flexible. As a result, the optimal inflation rate rises to around 3.5% to reduce further the costs from hitting the zero lower bound.

Since the 2008 financial crisis, a number of central banks have tackled the problem of the zero lower bound on inflation using a range of non-standard monetary

policies including quantitative easing, forward guidance, collateral adjustments, and actual negative interest rates. This demonstrates that the zero lower bound is not such a rigid constraint as theoretical economists have often assumed.

An important difference between South Africa and many other developed countries (including Europe, the USA, and Japan) is that, following the 2008 financial crisis, the South African Reserve Bank's interest rates did not fall to zero. In 2008 interest rates peaked at 12%, were subsequently reduced to 5% by the start of 2012, and remained at this level throughout the years following the financial crisis until they were further lowered at the start of the COVID-19 pandemic and reached a low of 3.5% in 2020 (see Fig. 2). Therefore, at no point during the last 20 years did the interest rate come close to the zero lower bound. This is exactly what the Andrade et al. (2019) model would have predicted with an inflation target range of 3–6% and where the performance of inflation would suggest that the realistic target was close to 5%. Figure 3 shows that the trend in real interest rates in South Africa has been for a very long-term fall since the mid-1990 s. Therefore, the South African experience has been similar to that of most of the rest of the world.

Given the volatility shown in Figs. 6 and 7 where, in particular, the standard error for interest rates has not gone above 0.8 since the early 2000 s with the start of inflation targeting and the interest rate itself has been around 5% or above over the same period, hitting the zero lower bound would involve interest rates moving by over 6 standard deviations, which is extremely unlikely. There is therefore some scope to lower the inflation target without incurring a serious risk of hitting the zero lower bound. It has also been widely demonstrated by the Bank of Japan, the European Central Bank, and the US Federal Reserve Board (Fed) that, even if the zero lower bound is hit, there are other policies which can take over. While the zero lower bound is often seen in the academic literature as a hard constraint in the real world, as these central banks have demonstrated, negative nominal interest rates are a possibility as is quantitative easing targeted at long rates and forward guidance.

3.6 Bias in the inflation measure

An additional argument for a non-zero inflation target is that there may be an inherent bias in the chosen measure of inflation. In the UK, for example, there is a well-known upward bias in the retail price index (RPI) which consistently overstates inflation relative to the CPI. This is due to a known error in the methodology used to calculate the RPI. Hence, the Bank of England targets the CPI. In a more general way it is accepted that no price index handles quality improvements in consumer goods well—consider the change in personal computers which has taken place in the last 30 years. Therefore, a positive rate of inflation should compensate for the failure to accurately measure quality improvements.

This is certainly an argument for having an inflation target, above zero. However, in the case of South Africa, where inflation has been close to 5% for a number of years, this is not relevant as, even if the inflation target were lowered, there would still be enough inflation to compensate for the quality effect.

4 Some operational issues in changing the target

There are a number of issues in inflation targeting which need to be considered, especially when thinking of changing the target. These are discussed in this section.

4.1 Inflation targeting versus price level targeting

If it was possible for a central bank to exactly hit its target then the issue of inflation or price level targeting would simply not matter. However, this is not possible and departures from the exact target will always occur. In this case the difference between the two is important. If a price level target is missed so that prices are above the target, then price level targeting would involve deflating the economy to get back to the original target. Inflation targeting would let bygones be bygones, ignore the previous mistake, and simply try to bring inflation back to its target. This, however, means greater uncertainty as to the long-term future price level, with obvious implications for the real value of future debt repayments, etc. This can lead to long-term price uncertainty, which can have detrimental effects for long-term planning.

The consensus in the literature is that under ‘normal’ circumstances inflation should fluctuate around the target or around the middle of the target if there is a range. However, there should be enough flexibility in the rules that in ‘exceptional’ times past inflation should be treated as a bygone. The recent burst of inflation which hit the international economy following the COVID-19 pandemic and the Ukraine war is an example of such an exceptional event, and no central bank is currently planning to bring the price level back to its pre-COVID-19 path.

4.2 The time horizon for inflation targeting

One important issue which has not so far been addressed is the horizon over which inflation targeting should be conducted. Sinclair (2011) gave an interesting example of the importance of this issue. Consider two countries, both of which have a 2% inflation target. One country oscillates between 3 and 1% in successive years while the other manages to achieve a constant 2.5%. The question is which country is doing better. In any one year the second country is closer to the target than the first and so would appear to be doing better. But if inflation is averaged over a number of years, then the first country will be spot on while the second country will exceed the target in every year. Over a number of years therefore, the price level of the second country will consistently rise faster than the target would imply, while the price level of the first country would remain close to the original target value. Most reasonable people would conclude that the first country is actually doing better.

The point about this simple example is that the horizon over which we assess the success in meeting the target is an important operational factor in setting up an inflation targeting regime. What should this horizon be? The answer to this is not simple and is essentially based on the structure of nominal contracts which take place in an economy. One purpose of an inflation target is to set price expectations for agents who are entering into nominal contracts. This could be in many

contexts including wage negotiations, commodity forward deals, and ordering and purchasing all manner of retail and wholesale goods. These are all reasonably short-term contracts for perhaps one or at most two years ahead. However, there are also very long-term contracts in terms of major investment projects both by the public and private sectors—for example, building a nuclear power station could take 20 or 30 years, and considerable uncertainty over the price level in that time frame could seriously impact the viability of such a project.

Clearly, an inflation target cannot be hit perfectly every year. It is therefore important that the targeting regime has both some short-term flexibility and a commitment that it will meet the target on average over a number of years. This does, however, mean that over time there will need to be periods of below target inflation and that the regime will begin to look like a price level targeting one. Woodford (2003) suggested some flexibility with an automatic penalty when inflation exceeds the target in the form of a lowering of the target to offset at least some of the excess inflation. This would help with the problem of anchoring expectations when the target is consistently missed. Consider our second country above. If inflation was consistently turning out to be 2.5% despite a 2% target inflation, expectations would eventually move towards 2.5%. Lowering the target itself in this circumstance might help to prevent this but at some cost to the transparency of the system. Posta and Tamborini (2022) developed a model where the European Central Bank has a central target of 2% but where there are symmetric upper and lower bands within which the bank would not intervene. This was shown to be welfare enhancing but only if there is no uncertainty as to the width of the bands.

This is essentially an argument for further flexibility in the system. Under ‘normal’ circumstances it might be sensible to hit an average inflation target over two to three years. However, when unusual events such as the COVID-19 pandemic or the Ukraine war happen, it might be both economically and politically infeasible to try and offset the inflationary shock over the following four or five years. Transparency would require that the central bank makes it clear when it is treating inflation as a bygone and to explain why a particular event is exceptional. It is certainly true that none of the economies that have experienced the surge in inflation since 2020 are attempting to return the price level to its former path, which would entail a considerable deflation.

4.3 Matching the inflation target of trading partners

An interesting argument for adopting something close to the general 2% target is simply to adopt what everyone else does. Setting an inflation target well above other economies (especially those of the developed world) would imply a need for a continual devaluation of the exchange rate against the world’s major currencies. In particular, if uncovered interest parity does not hold perfectly, then targeting a higher rate than in other countries can lead to distortions in either relative interest rates or capital flows, which may adversely affect the economy (Eitrhein and Qvigstad 2020).

So, to take the case of South Africa, if the developed world is generally targeting a 2% inflation rate reasonably successfully and South Africa is successfully targeting 5%, then we would expect the rand to devalue by 3% a year over the long run to maintain trade competitiveness. The open arbitrage condition would suggest that South African interest rates should be 3% higher than in other countries. If this did not happen, there would be international competitiveness effects and capital market distortions. In fact, since 2005, the average rate of depreciation for the rand was 5% a year, which is very close to the actual average inflation rate. However, as shown in Fig. 5 the actual change in the exchange rate has been very erratic with occasional devaluations of up to 40% in one year, but there have also been many periods of appreciation, in one case by up to 30% in a year. This level of instability and uncertainty is bound to have serious effects on long-term trade.

4.4 Experience of changing the target

Given the large number of countries which have adopted inflation targeting, it is surprising that relatively few changes in the initial target have come about. There are, however, a number of instances of changes in the target rate. These mostly entail lowering the rate but there are a few which have raised it slightly. Some examples of lowering the target are:

- Chile, which adopted targeting in 1990 after a period of around 20% inflation, then gradually lowered the target rate to 5% by 1999, and then further lowered the target to $3 \pm 1\%$ by 2007;
- Thailand, where the central bank moved from a target of $2.5 \pm 1.5\%$ to a 1–3% band in 2020, thus basically narrowing the band around 2%;
- The UK, where the Bank of England changed from targeting the RPI to the CPI in 2003 and at the same time lowered the target from 2.5 to 2%. This actually reflects a mistake in the calculation of the RPI which consistently overstates inflation compared with the CPI in the UK;
- Norway, where the central bank lowered its target from 2.5 to 2.0% in 2018; and
- Korea, where the Bank of Korea had a target of $3 \pm 1\%$ in 2013 and gradually reduced this to a 2% target by 2016.

There have been no substantial instances of an increase in the target, although there has been some redesigning of the target. In 1990 New Zealand had a 0–2% target range, which it gradually moved to a 1–3% range by 2002. This was essentially homing in on the dominant 2% target. The Bank of Japan and the European Central Bank also changed their target from below 2% to a simple 2% target, again essentially coming into line with most other central banks. None of these changes had any discernible consequences in terms of output or unemployment effects.

Nonetheless, an important issue is to ask whether there is a cost to changing the target rate of inflation and if so, how it can be mitigated. This is essentially asking whether there is a short-, medium-, and long-term trade-off between inflation and

unemployment. There is a clear and extensive agreement that the long-run Phillips curve is vertical so that there should be no long-run effect on unemployment from the change in the level of inflation, but it is possible that during a transition period there could be adverse consequences. Figure 4 shows the unemployment rate for South Africa from 2002 to 2024. Unemployment fell steadily until 2007 and then there was a steady rise until it peaked in 2022 at close to 34%, a very high rate. There is little obvious relationship between unemployment and the inflation rate, as shown in Fig. 1. Traditionally, inflation is determined by a combination of expected inflation and economic activity, usually captured by unemployment. The crucial question to ask here is what the relative importance of these two effects are.

There is a substantial literature on estimating Phillips curves for South Africa. Hodge (2002) undertook an early study of the Phillips curve for South Africa and found evidence of a significant trade-off between inflation and unemployment in a fairly simple econometric framework. Umaru et al. (2013), however, argued that there was no causal link between unemployment and inflation. This is important as it implies that raising unemployment is not the way to reduce inflation, which is reasonable given the very high levels of unemployment in South Africa, which would have implied a considerable fall in inflation if the Phillips curve effect was both causal and powerful. Vermeulen (2015) argued that, while inflation harmed output and unemployment, there were no significant reverse effects from unemployment on inflation. Later work used more sophisticated econometric techniques, which emphasizes the difficulty in finding robust evidence of a convincing effect of unemployment on inflation. Nasr et al. (2015) used a four-state Markov-switching vector autoregressive model (MS-VAR) to find some effects, but this was very regime dependent. Vermuelen (2017) found evidence again for an effect of inflation on unemployment but no effect in the other direction. Kabundi et al. (2015, 2019) did an extensive study and found that there was considerable time variation in the parameters of the Phillips curve and that it had been consistently flattening since the early 2000 s. Buthelezi (2023) used a Hamilton-switching model and also found that a sophisticated model was needed to find any support for the Phillips curve. Finally, Dladla and Malikane (2022) stated that ‘the literature on the Phillips curve in South Africa finds little evidence of the effect of demand pressure on the inflation rate’, although they did go on to argue that this was due to misspecification.

We estimate two versions of the Phillips curve for South Africa using an errors in variable measure of expectations and estimating the model using generalized methods of moments (GMM) over the period 2002 month 1–2023 month 5. The result for the first equation, which is in the traditional Phillips curve form, is:

$$\inf_t^e = 0.03 \inf_t^e + 0.14unemp_t - 0.18unemp_{t-1} + 0.04unemp_{t-2} + 1.62 \inf_{t-1} - 0.65 \inf_{t-2} \quad (1)$$

(1.2) (2.4) (1.6) (0.5) (12.9) (5.3)

(HAC robust ‘t’ statistics in parentheses).

Weak instrument test Crag–Donald F-Stat 5.6

Endogeneity test 1.96 (p value 0.37).

Normality BJ test 9.4 (p value 0.008).

P-value of the J statistic 0.06.

DW statistic 2.5

R^2 0.96

where $\inf_t$ is the monthly 12-month inflation rate at time t , $\inf_t^e$ is the expected inflation rate 12 months ahead at time t , and $unemp_t$ is the percentage unemployment rate at time t . Student 't' statistics are in parentheses. The long-run coefficient on expected inflation is given by $0.03/(1 - 1.62 + 0.65) = 1$ and the unemployment rate effect is positive, but the dynamic effects add up to a positive change, which confirms no long-run effect from unemployment but, if anything, a temporary positive effect, which is the opposite effect to the standard short-run Phillips curve.

The second equation is in the form of an autoregressive distributed lag (ARDL) error correction model:

$$\Delta \inf_t = -0.03(\inf_{t-1} - 1.18 \inf_{t-1}^e + 0.03unemp_{t-1}) + 0.62\Delta \inf_{t-1} \quad (2)$$

(3.8) (1.9) (0.3) (5.9)

(HAC robust 't' statistics in parentheses).

Normality BJ test 11.35 (p value 0.003).

P-value of the J statistic 0.11.

DW statistic 2.4

R^2 0.3

Where Δ is the first difference operator. The long-run solution for inflation is $\inf = 1.18 \inf^e - 0.03unemp$ where the unemployment is again highly insignificant although now correctly signed. The error correction model (ECM) adjustment coefficient of -0.03 suggests a half-life of around 30 months, which implies a relatively slow adjustment of actual inflation to expected inflation.

In some ways this discussion is a side issue from the main purpose of this paper. The key point to take from both the literature and these results is that expectations are a much more significant determinant of inflation than economic activity or unemployment. The whole point of inflation targeting and forward guidance is to affect expectations directly. It is therefore not necessary to reduce inflation through raising unemployment. Even if this channel was significant and correctly signed, the appropriate way to reduce inflation in an inflation targeting regime is for a credible central bank to reduce the expected rate of inflation.

These changes suggest two conclusions. First, and most importantly, downward adjustments should be made progressively and over quite a long period of time (three to four years given the half-life calculation above). Too rapid adjustment may involve missing the target and a consequent loss of credibility. Second, it is possible through gradual reduction to substantially reduce inflation over time in a stable way and with little or no consequences for economic activity.

5 Conclusions

There is a clear distinction between the academic literature and the judgement of practical central bankers over the optimal target for inflation. The majority of the academic literature, especially the earlier literature, recommends a zero inflation target, but no central bank has set such a target. The majority of developed economies have chosen a 2% target and many developing economies have chosen something rather higher (see Appendix B). I take the view that if a theoretical result does not smell right to an experienced practitioner, then it is probably wrong. The early theoretical literature abstracted from a number of real-world features which, we now know, are crucial to obtaining a reasonable estimate of the optimal inflation target. These include:

1. The benefits to growth of low inflation. Low inflation as opposed to high inflation is undoubtedly beneficial to growth, but, given the available data this is not an argument for zero inflation and a rate of 2–3% inflation would probably attract all the benefits available in almost all economies, including South Africa.
2. Downward rigidity in price setting. It is an accepted stylized fact that at least some prices are downwardly rigid, most notably nominal wages. Maintaining an optimal set of relative prices is therefore made easier if there is a moderate rate of inflation. Academic studies which allow for this effect find an optimal rate quite close to 2% for developed European economies and, given the structural difference between Europe and South Africa, it seems reasonable to suggest that this effect would argue for a somewhat higher rate for South Africa of perhaps 3–4%.
3. The zero lower bound. The academic literature has devoted a lot of time to this effect and a broad consensus would be that a target rate of around 2% is optimal for developed economies such as the USA and Europe. South Africa has been following a target which seems, in practice, to be around 5% and has not even been close to hitting the zero lower bound for interest rates. This experience confirms the finding of these models that at a 5% target the chances of hitting the zero lower bound are negligible. This is confirmed by the conditional standard error for nominal interest rates (Appendix Figs. 6 and 7) which suggests that a two-standard deviation movement downwards in interest rates would not have caused interest rates to become zero at any point in recent history. It is also important to remember that central banks that have experienced the zero lower bound have found other ways to operate an effective monetary policy, including negative interest rates, forward guidance, and quantitative easing. On this basis there does seem room for South Africa to lower its target inflation rate somewhat.
4. Government debt servicing and an incomplete tax system. Any level of inflation acts as a tax on the economy. It is, however, a regressive tax and does not affect all members of society equally. Wealthy individuals may be able to manage their wealth in a way which largely avoids this tax. Housing wealth, where house prices are expected to rise in line with inflation, or investments in the stock market, which should also rise in line with inflation, are just some ways in which the

wealthy can avoid the inflation tax. The tax therefore falls most heavily on the poorer members of society and does not constitute a good way to raise taxes.

5. Bias in the inflation rate. Most statistical offices accept that there is some upward bias in measured inflation, if only because of quality improvements in goods over time. If, when the true inflation rate is zero and measured inflation is, say, 2%, then targeting a zero inflation rate would actually be targeting -2% deflation. Hence a 2% target would actually be targeting a 0% actual inflation rate. Measuring the extent of bias in any economy is virtually impossible. If it was possible then statistical offices would correct the bias. Therefore, it is hard to say whether the bias in South Africa is greater or less than in other countries. However, it is almost certainly true that a 2–3% target would allow for this bias.

The obvious point to make at this stage is that each of these arguments may suggest a different optimal rate of inflation but they are not additive. Therefore, if the zero lower bound inflation target was 3% and the optimal rate to deal with downward rigidities was 4%, then the choice should be 4% not 7%, as 4% will also eliminate the chance of hitting the zero lower bound.

Taken together, all these arguments seem to suggest that the lower end of the inflation target range in South Africa should be lowered so that the range should at least include 2% and that gradually, over time, actual inflation should be brought to the lower end of the range. Given the experience of other central banks, this needs to be done quite gradually and over a long time horizon, perhaps five or more years, with clear and transparent announcements as to what is planned. If this is done gradually and with clear transparency, the expectation effects on inflation should be sufficient to avoid any need for an increase in inflation.

Appendix A: Basic data and its variability

This appendix begins by showing the basic data for the CPI inflation rate for South Africa and the Central Bank interest rate, both expressed as an annual rate using monthly data.

Central Bank Interest Rate—Federal Reserve economic data, Federal Reserve Bank of St Louis, IRSTCB01ZAM156 N, Interest Rates: Immediate Rates (< 24 Hours): Central Bank Rates: Total for South Africa, Percent, Monthly, Not Seasonally Adjusted.

Inflation Rate—Federal Reserve economic data, Federal Reserve Bank of St Louis, CPALTT01ZAM659 N, Consumer Price Indices (CPIs, HICPs), COICOP 1999: Consumer Price Index: Total for South Africa, Growth rate same period previous year, Monthly, Not Seasonally Adjusted.

Unemployment rate—Federal Reserve economic data, Federal Reserve Bank of St Louis, LRUN64 TTZAQ156S. This data is semi annual from 2000 to 08 when it became quarterly. The data was then interpolated to monthly data using a cubic spline.

See Appendix Figs. 1, 2, 3, 4, 5, 6, 7 and Tables 1, 2, 3, 4, 5.

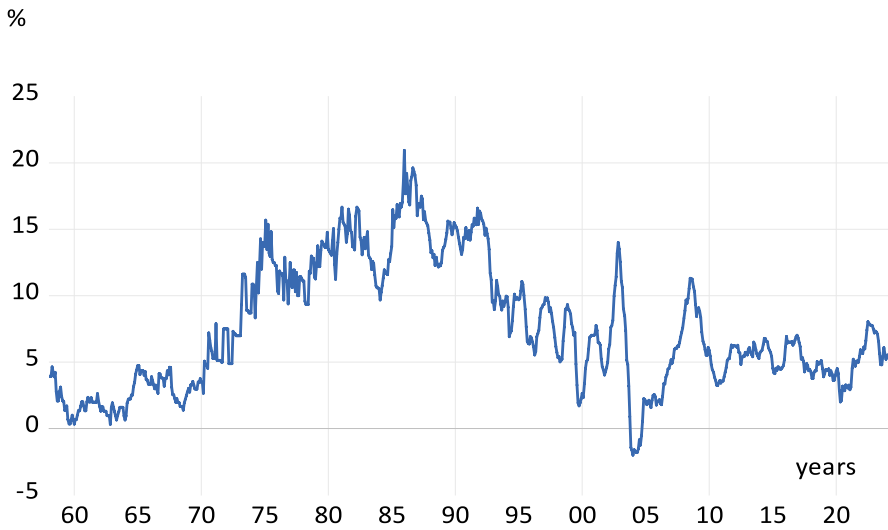


Fig. 1 Annual CPI inflation. Source: author's calculations

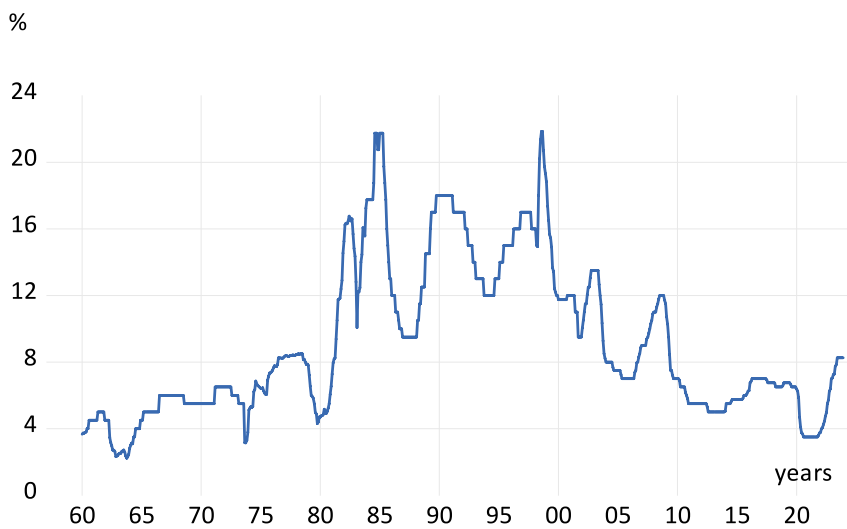


Fig. 2 The central bank interest rate. Source: author's calculations

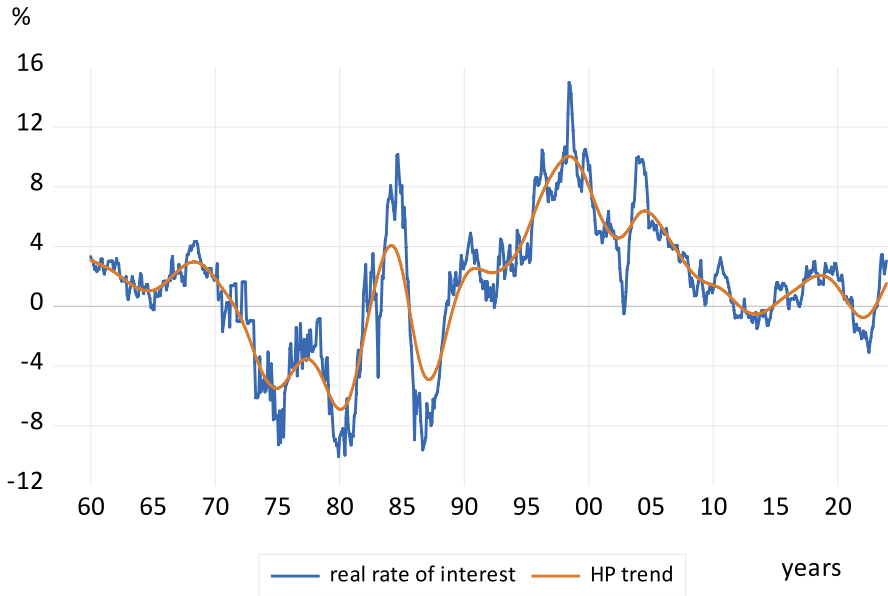


Fig. 3 The real rate of interest and an HP trend. Source: author's calculations

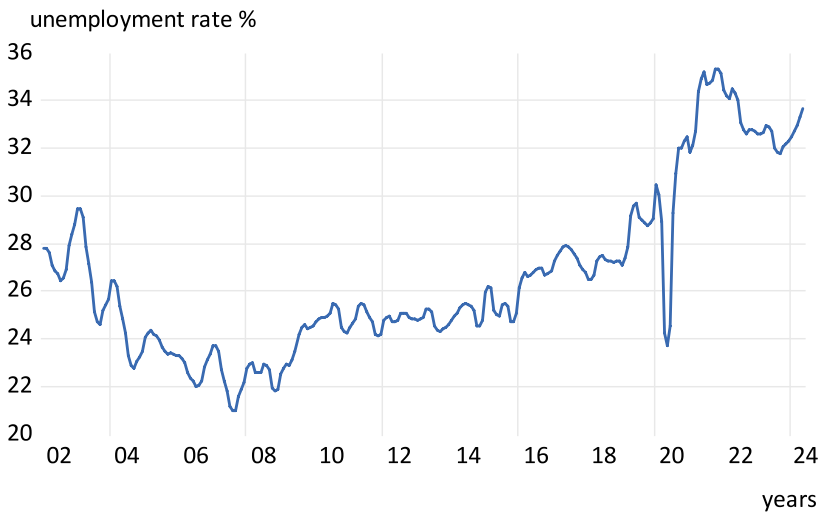


Fig. 4 The unemployment rate. Source: author's calculations

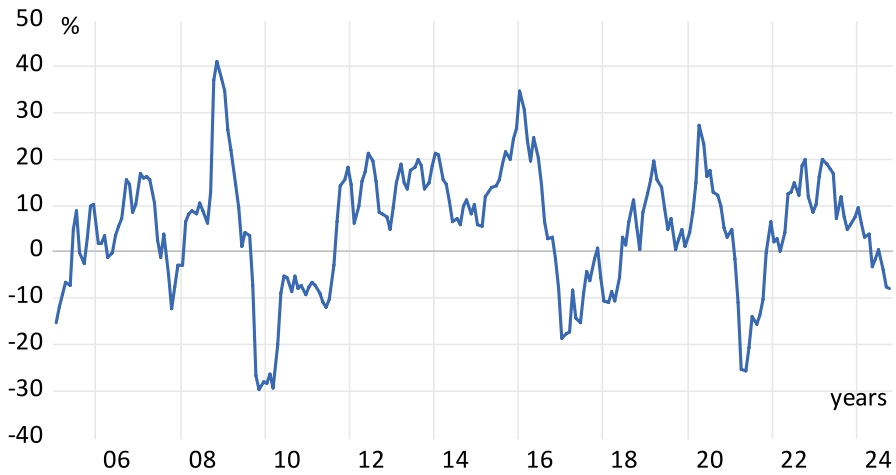
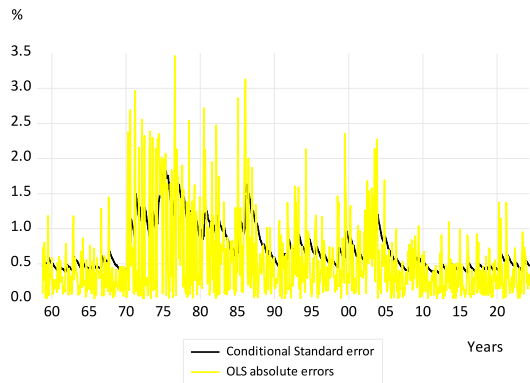


Fig. 5 Rate of rand devaluation. Source: author's calculations

Fig. 6 The conditional standard error and the absolute value of the OLS residuals for inflation. Source: author's calculations



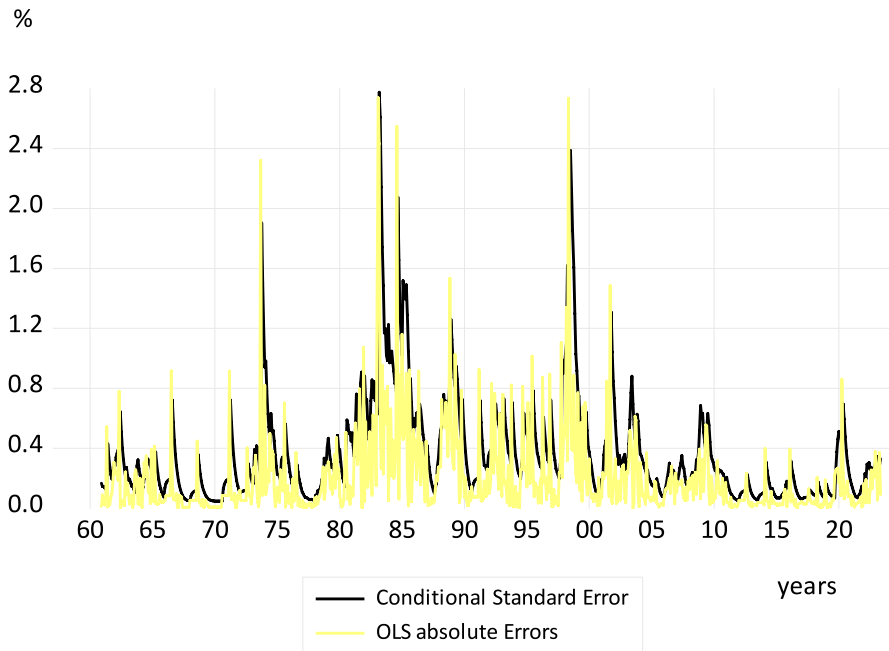


Fig. 7 The conditional standard error and the absolute value of the OLS residuals for the interest rate.
Source: author's calculations

Table 1 Summary statistics.
Source: author's calculations

	Interest rate	Inflation
Mean	9.068912	7.744094
Median	7.087857	6.558572
Maximum	21.85000	20.94241
Minimum	2.220000	– 1.999269
Std. Dev	4.621742	4.738477
Skewness	0.818141	0.408597
Kurtosis	2.645344	2.214629
Jarque–Bera	89.70243	41.10762
Probability	0.000000	0.000000
Sum	6964.924	5947.464
Sum Sq. Dev	16,383.50	17,221.58
Observations	768	768

Table 2 OLS results for inflation. *Source:* author's calculations

Dependent variable: Inflation				
Method: Least squares				
Date: 08/13/24 Time: 09:47				
Sample (adjusted): 1958M10 2024M05				
Included observations: 788 after adjustments				
Variable	Coefficient	Std. error	t-Statistic	Prob
C	0.139460	0.053165	2.623161	0.0089
Inflation(− 1)	0.972817	0.035331	27.53433	0.0000
Inflation(− 2)	0.054604	0.049481	1.103546	0.2701
Inflation(− 3)	0.063355	0.049356	1.283640	0.1996
Inflation(− 4)	− 0.133498	0.041820	− 3.192172	0.0015
Inflation(− 7)	0.114771	0.033474	3.428665	0.0006
Inflation(− 9)	− 0.089956	0.025233	− 3.565026	0.0004
R-squared	0.973673	Mean dependent var		7.609869
Adjusted R-squared	0.973471	S.D. dependent var		4.759999
S.E. of regression	0.775297	Akaike info criterion		2.337702
Sum squared resid	469.4478	Schwarz criterion		2.379183
Log likelihood	− 914.0548	Hannan-Quinn criter		2.353649
F-statistic	4814.089	Durbin-Watson stat		2.002514
Prob(F-statistic)	0.000000			

Table 3 OLS results for the central bank interest rate. *Source:* author's calculations

Dependent variable: Interest Rate				
Method: Least squares				
Date: 08/13/24 Time: 09:56				
Sample (adjusted): 1960M12 2023M05				
Included observations: 750 after adjustments				
Variable	Coefficient	Std. error	t-Statistic	Prob
C	− 0.018087	0.031514	− 0.573944	0.5662
Interest rate(− 1)	1.105746	0.036062	30.66209	0.0000
Interest rate(− 2)	− 0.131772	0.044455	− 2.964139	0.0031
Interest rate(− 4)	− 0.111471	0.044508	− 2.504489	0.0125
Interest rate(− 5)	0.152894	0.053877	2.837835	0.0047
Interest rate(− 6)	− 0.103872	0.044377	− 2.340676	0.0195
Interest rate(7)	0.091661	0.007299	12.55829	0.0000
Interest rate(− 8)	0.091328	0.044618	2.046867	0.0410
Interest rate(− 9)	− 0.160670	0.044332	− 3.624231	0.0003
Interest rate(− 11)	0.068073	0.020284	3.355975	0.0008
R-squared	0.993730	Mean dependent var		9.150005
Adjusted R-squared	0.993653	S.D. dependent var		4.635826
S.E. of regression	0.369320	Akaike info criterion		0.858937
Sum squared resid	100.9340	Schwarz criterion		0.920538
Log likelihood	− 312.1014	Hannan-Quinn criter		0.882673
F-statistic	13,030.35	Durbin-Watson stat		2.047948
Prob(F-statistic)	0.000000			

Table 4 A GARCH model for inflation. *Source:* author's calculations

Dependent variable: Inflation

Method: ML ARCH—Normal distribution (BFGS/Marquardt steps)

Date: 08/12/24 Time: 13:51

Sample (adjusted): 1958M10 2024M05

Included observations: 788 after adjustments

Convergence achieved after 56 iterations

Coefficient covariance computed using outer product of gradients

Presample variance: backcast (parameter = 0.7)

GARCH = C(8) + C(9)*RESID(-1)^2 + C(10)*GARCH(-1)

Variable	Coefficient	Std. error	z-Statistic	Prob
C	0.129135	0.036082	3.578937	0.0003
Inflation(− 1)	1.144503	0.038397	29.80698	0.0000
Inflation(− 2)	− 0.144827	0.062838	− 2.304761	0.0212
Inflation(− 3)	0.051562	0.063932	0.806520	0.4199
Inflation(− 4)	− 0.058504	0.050583	− 1.156601	0.2474
Inflation(− 7)	0.043293	0.033477	1.293215	0.1959
Inflation(− 9)	− 0.055583	0.023504	− 2.364793	0.0180
Variance equation				
C	0.007889	0.002312	3.412338	0.0006
Resid(− 1)^2	0.096789	0.017940	5.395228	0.0000
Garch(− 1)	0.891888	0.017092	52.18313	0.0000
R-squared	0.972647	Mean dependent var		7.609869
Adjusted R-squared	0.972437	S.D. dependent var		4.759999
S.E. of regression	0.790262	Akaike info criterion		2.033135
Sum squared resid	487.7448	Schwarz criterion		2.092392
Log likelihood	− 791.0551	Hannan-Quinn criter		2.055915
Durbin-Watson stat	2.337213			

Table 5 A GARCH model for the central bank interest rate. *Source:* author's calculations

Dependent variable: Interest Rate

Method: ML ARCH—Normal distribution (BFGS/Marquardt steps)

Date: 08/12/24 Time: 14:29

Sample (adjusted): 1960M12 2023M05

Included observations: 750 after adjustments

Convergence achieved after 88 iterations

Coefficient covariance computed using outer product of gradients

Presample variance: backcast (parameter = 0.7)

GARCH = C(11) + C(12)*RESID(-1)^2 + C(13)*GARCH(-1)

Variable	Coefficient	Std. error	z-Statistic	Prob
C	− 0.011226	0.005772	− 1.945001	0.0518
Interest rate(− 1)	0.942438	0.059475	15.84599	0.0000
Interest rate(− 2)	− 0.050282	0.083539	− 0.601901	0.5472
Interest rate(− 4)	− 0.063423	0.077189	− 0.821660	0.4113
Interest rate(− 5)	0.080419	0.083748	0.960248	0.3369
Interest rate(− 6)	− 0.045593	0.061361	− 0.743023	0.4575
Interest rate(7)	0.151904	0.004739	32.05192	0.0000
Interest rate(− 8)	0.023389	0.042645	0.548460	0.5834
Interest rate(− 9)	− 0.036251	0.036644	− 0.989283	0.3225
Interest rate(− 11)	− 0.001822	0.014305	− 0.127357	0.8987
Variance equation				
C	0.000807	0.000154	5.226337	0.0000
Resid(− 1)^2	0.679170	0.080258	8.462327	0.0000
Garch(− 1)	0.619053	0.022217	27.86411	0.0000
R-squared	0.992990	Mean dependent var		9.150005
Adjusted R-squared	0.992905	S.D. dependent var		4.635826
S.E. of regression	0.390480	Akaike info criterion		0.184268
Sum squared resid	112.8312	Schwarz criterion		0.264349
Log likelihood	− 56.10046	Hannan-Quinn criter		0.215125
Durbin-Watson stat	1.636825			

Appendix B

See Appendix Table 6.

Table 6 Inflation targeting regimes in 2022. *Source:* Central Bank News <https://www.centralbanknews.info/p/inflation-targets.html>

Country	Central bank	Target (%)
Albania	Bank of Albania	3.00 ± 1
Argentina ³	Central Bank Rep. Argentina	0 mon. base
Armenia	Central Bank of Armenia	4.00 ± 1.5
Australia	Reserve Bank of Australia	2.00–3.00
Azerbaijan	C.B. of Rep. of Azerbaijan	4.00 ± 2.0
Bangladesh ¹⁰	Bangladesh Bank	5.30
Belarus ¹⁹	N.B. of Rep. of Belarus	5.00
Botswana	Bank of Botswana	3.00–6.00
Brazil ¹²	Central Bank of Brazil	3.75 ± 1.5
Canada	Bank of Canada	2.00 ± 1.0
Chile	Central Bank of Chile	3.00 ± 1.0
China ⁵	People's Bank of China	around 3.00
Colombia	Central Bank of Colombia	3.00 ± 1.0
Dem. Rep. Congo	Central Bank of Congo	7.00
Costa Rica	Central Bank of Costa Rica	3.00 ± 1.0
Czech Republic	Czech National Bank	2.00 ± 1.0
Dominican Rep	C.B. of Dominican Repl	4.00 ± 1
Egypt ⁷	Central Bank of Egypt	7.00 ± 2
Eswatini	Central Bank of Eswatini	3.00–7.00
Euro Area ¹	European Central Bank	2.00
Gambia	Central Bank of the Gambia	5.00
Georgia	National Bank of Georgia	3.00
Ghana	Bank of Ghana	8.00 ± 2.0
Guatemala	Bank of Guatemala	4.00 ± 1.0
Hungary	Central Bank of Hungary	3.00 ± 1.0
Honduras	Central Bank of Honduras	4.00 ± 1.0
Iceland	Central Bank of Iceland	2.50
India	Reserve Bank of India	4.00 ± 2.0
Indonesia	Bank Indonesia	3.00 ± 1.0
Israel	Bank of Israel	1.00–3.00
Jamaica	Bank of Jamaica	4.0–6.0
Japan	Bank of Japan	2.00
Kazakhstan	National Bank of Kazakhstan	4.00–6.00
Kenya	Central Bank of Kenya	5.00 ± 2.50
Kyrgyzstan	N.B. of Kyrgyz Republic	5.00–7.00
Liberia ²	Central Bank of Liberia	8.50/+ -2
Malawi	Resserve Bank of Malawi	5.00
Mexico	Bank of Mexico	3.00 ± 1.0
Moldova	National Bank of Moldova	5.00 ± 1.5
Mongolia	Bank of Mongolia	6.00 ± 2
Mozambique	Bank of Mozambique	5.60

Table 6 (continued)

Country	Central bank	Target (%)
Nepal	Nepal Rastra Bank	6.00
New Zealand	Res. Bank of New Zealand	2.00 $\pm$ 1.0
Nigeria	Central Bank of Nigeria	6.00–9.00
Norway	Norges Bank	2.00
Pakistan	State Bank of Pakistan	6.00
Paraguay	Central Bank of Paraguay	4.00 $\pm$ 2.0
Peru	Central Reserve Bank of Peru	2.00 $\pm$ 1
Philippines	Bangko Sentral ng Pilipinas	3.00 $\pm$ 1.0
Poland	National Bank of Poland	2.50 $\pm$ 1.0
Romania	National Bank of Romania	2.5 $\pm$ 1.0
Russia	Bank of Russia	4.00
Rwanda	National Bank of Rwanda	5.00 $\pm$ 3
Samoa	Central Bank of Samoa	3.00
Serbia ¹⁷	National Bank of Serbia	3.00 $\pm$ 1.5
South Africa	South African Reserve Bank	3.00–6.0
South Korea	Bank of Korea	2.00
Sri Lanka	Central Bank of Sri Lanka	4.00–6.00
Sweden	The Riksbank	2.00
Switzerland	Swiss National Bank	< 2.00
Tajikistan	N. B. of Tajikistan	6.0 $\pm$ 2.0
Tanzania	Bank of Tanzania	5.00
Thailand	Bank of Thailand	1.00–3.00
Tonga	National Res. Bank of Tonga	5.00
Turkey ¹⁸	C.B. of Rep. of Turkey	5.00 $\pm$ 2
Uganda	Bank of Uganda	5.00 $\pm$ 2.0
Ukraine	National Bank of Ukraine	5.00 $\pm$ 1
United Kingdom	Bank of England	2.00
Uruguay	Central Bank of Uruguay	3.00–7.00
Usa ¹⁶	Federal Reserve	2.00
Uzbekistan ¹⁵	C. B. of Rep. Of Uzbekistan	5.00
Vietnam ⁹	State Bank of Vietnam	< 4
West African States	Central Bank of West African States	2.00 $\pm$ 1
Zambia	Bank of Zambia	6.0–8.0

¹Symmetric medium-term target vs close to but below 2.0%; ²Q3 2021 forecast; ³Zero growth in monetary base as of Sep. 26, 2018; ⁵CCP National Congress, unchanged since 2015; ⁷on average in Q4 2022; ⁹14% credit growth; ¹⁰5.4% for FY20-21, 5.3% for FY 21–22, 5.2% FY 22–23; ¹²3.75% in 2021, 3.5% in 2022, all $\pm$ 1.5 pp; ¹³5.0% in 2023, 10% 2021; ¹⁶August 2020 Fed adopts flexible form of inflation targeting that seeks inflation that averages 2% over time; ¹⁷Until 2023; ¹⁸9.4% inflation by end-2021, 5% in 2023; ¹⁹end-2021

Declarations

Conflict of interest There is no conflict of interest, there are no ethical issues raised by this work, no individuals are involved in this work so no consent is required. All data is available from the Federal reserve database ‘FRED’. This work was funded by the UN-Wider University.

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